

chapter # :-

Web mining & Text miningWeb mining:

The use of data mining techniques to discover and extract information from web documents and services.

Challenge:

The web is massive and growing daily, with semi-structured or unstructured content, making traditional database management and information retrieval solutions inadequate.

Four subtasks of web-mining(i) Resource finding:

Retrieving data from online or off line multimedia sources, such as news, articles, forums, blogs and text content from HTML documents after removing HTML tags.

(ii) Information selection and preprocessing
 Transforming retrieved data through preprocessing steps like removing stop words and stemming, or representing the text in a desired format such as finding phrases or using first order logic.

(iii) Generalization

Discovering general patterns within individual websites and across multiple sites using ML and DM techniques.

(iv) Analysis

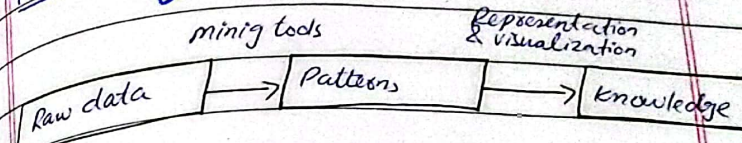
Validating and interpreting mined patterns.

Types of web Tools

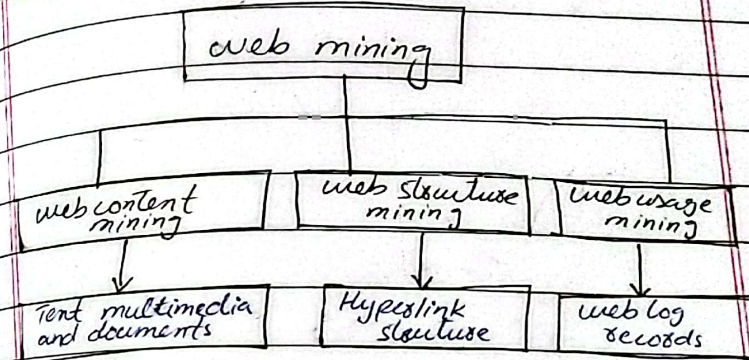
Usex-navigation
Tools

offline analysis
Tools

Process of web mining:



web mining Types



Applications of web mining:

- o improve the power of web search engine
- o used for web searching
- o Predict user behavior

HITS algorithm

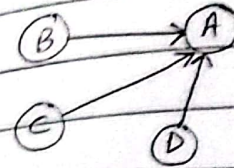
Hypertext induced Topic Search (HITS)

- ⇒ Link analysis algorithm that rates web pages.
- ⇒ was developed by John Kleinberg in 1999.
- ⇒ Used in web search engines.
- ⇒ Helps identify authoritative pages on the web.
- ⇒ Also finds pages of high quality or authority.
- ⇒ HITS uses hubs and relationships b/w web pages.
- ⇒ The web has two important types of pages:

Pages with the best information on a topic.

Node that is pointed by many hubs.

an authority is a node that contains useful and relevant information that is linked to by many other nodes.



HITS algorithm steps:

Sampling Component:

- Initialize the hub and authority of each node with a value of 1.
- For each iteration, update the hub and authority of every node in the graph.
- The new authority is the sum of the hubs of its parents.
- The new hub is the sum of the authority of its children.

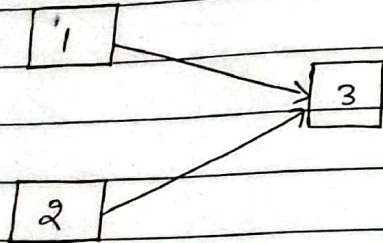
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- o Normalize the new authority and hub.
- weight-propagation components.
- o Calculate the hub and authority scores through an iterative process.

Example #:

Compute the hub and authority weights for the following graph:



- o find the adjacency matrix of the graph

A =

	1	2	3
1	0	0	1
2	0	0	1
3	0	0	0

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o Transpose of matrix:

$$A^t = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

Assume the initial hub weight vector is $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

compute the authority weight vector

$$V = A^t \cdot U$$

$$V = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

update the hub weight:

$$U = A \cdot V$$

$$U = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}$$

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Results:

hub weights

$$\begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}$$

authority weights

$$\begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$



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LOGOSOM Algorithm

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Log based self-organizing maps.
The LOGOSOM algorithm uses user behavior to organize web pages.

How it works?

Challenge with web searches:

- As the web grows, finding relevant information becomes harder.
- Traditional search results often include many irrelevant pages.
- User behavior:
- HITS uses static web structure
- LOGOSOM uses dynamic information about how users navigate the web.

Self-organizing maps (SOM):

- SOMs group data into clusters and provide a graphical representation of the relationships among these clusters.

LOGOSOM organizes web pages into 2D table.

Process:

- Start with a web log file that records user activity.
- Define transactions as sets of user-requested web pages.
- Represent these transactions in a binary matrix, where rows are transactions and columns are URLs.

Data reduction:

Use k-mean for reduction.

Applications:

- Companies use.
- Understanding which web pages user frequently visit together.

Define path-Traversal pattern mining?

Path-Traversal pattern mining involves capturing user access patterns in a web environment.

This technique focuses on the sequence of pages user visit, rather than just individual page views.

Extracting meaningful patterns from these sequences can be challenging.

What is Text mining?

Text mining extracts useful information from large collection of textual data, leveraging techniques from text analysis and internet search.

⇒ This process handles semi-structured text found in electronic publications and web.

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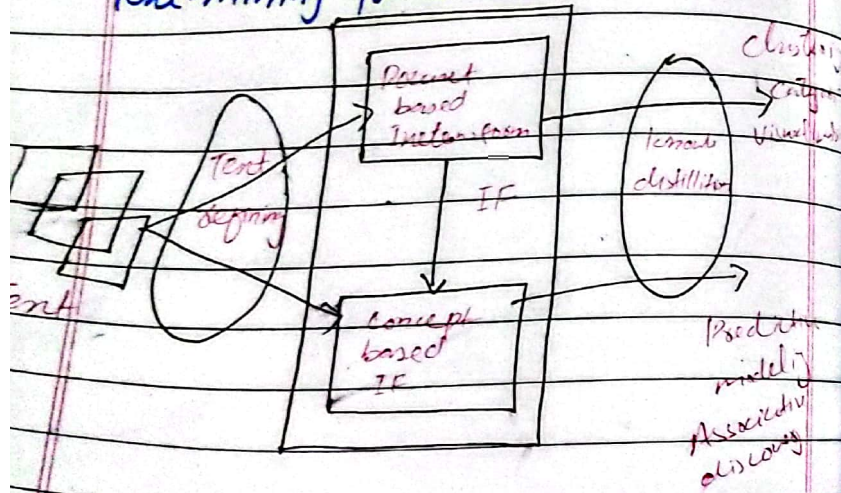
⇒ Key steps:

⇒ Transforming text into structured immediate forms and then distilling knowledge to find patterns, categorize content and visualize relationships.

⇒ Applications:

- market research
- Business intelligence

Text-mining framework:



PageRank Algorithm:

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Created by Sergey Brin and Larry Page, cofounder of Google.

Contributed to Google's early success.

Provides a global ranking of web pages based on their authority.

Same like HITS algorithm

Concept:

◦ Each link from page a to page b acts as vote by a for b.

◦ Votes are weighted by the PageRank score of the originating page.

Formula:

$$Pr(u) = \frac{1-d + d \sum_{v \in \text{out}(u)} Pr(v)}{N}$$

Q. What is LSA?

⇒ Latent semantic analysis
LSA improves information retrieval by focusing on the semantic meaning of words rather than simple string matching.

⇒ it addresses issues with synonyms & polysemes using SVD to reduce data dimensions.

⇒ This process captures key patterns and semantic similarities among documents, enhancing tasks like clustering & classification.
